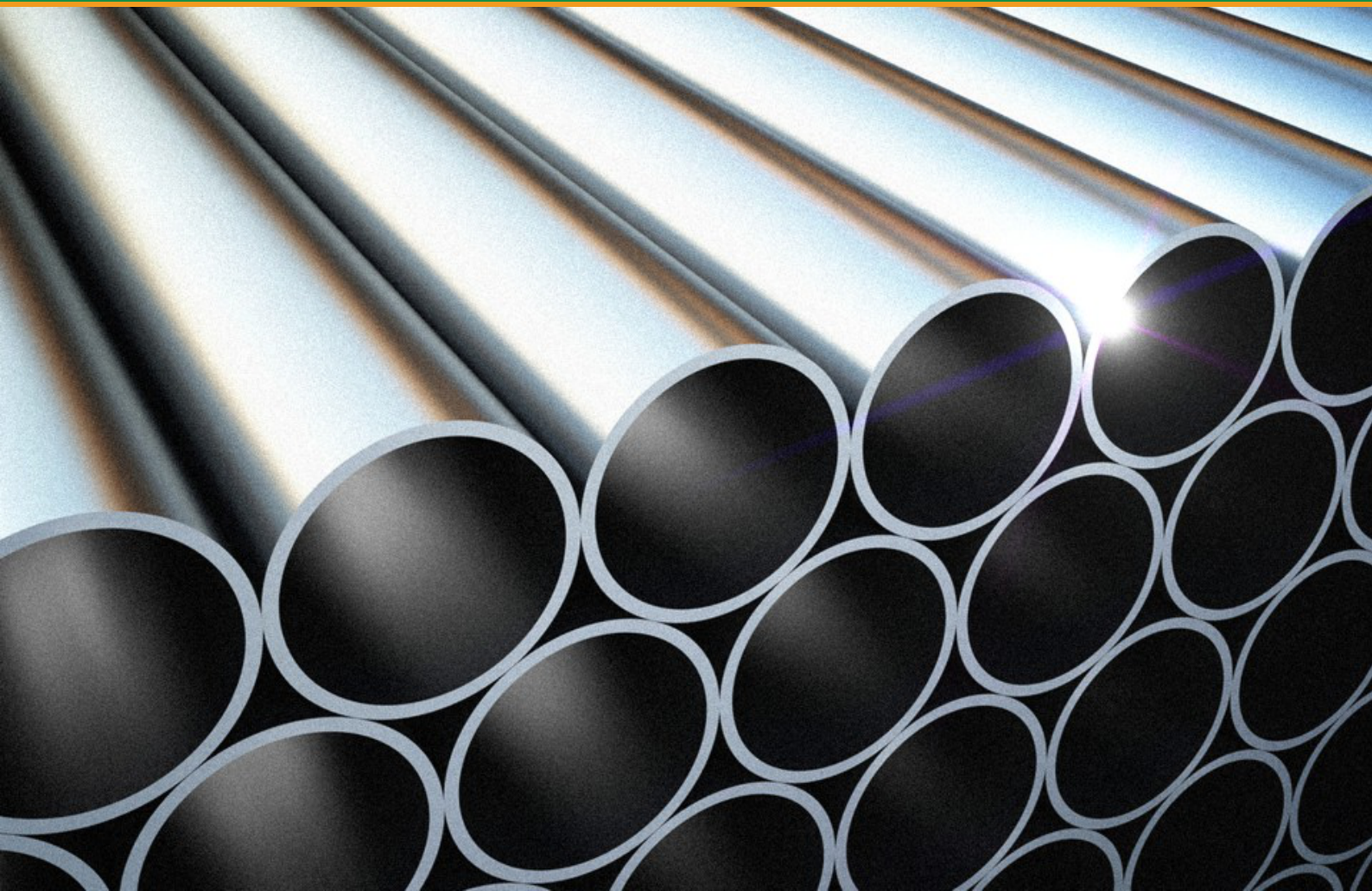
PROCESS PIPE

TUBES CATALOGUE



STRUCTURAL TUBE

MILD STEEL DIMENTIONS AND MASS

ROUNDS

Outside diameter mm	Wall Thickness mm										
	1.6	2.0	2.2	2.5	2.8	3.0	3.5	4.0	4.5	5.0	6.0
19.05	0.725	0.843	-	-	-	-	-	-	-	-	-
21.40	-	0.951	-	1.158	-	1.364	-	-	-	-	-
25.40	0.938	1.153	-	1.411	-	-	-	-	-	-	-
26.90	-	1.227	1.339	1.503	-	-	-	-	-	-	-
31.80	1.198	1.478	-	1.817	-	2.129	-	-	-	-	-
34.10	-	1.562	1.708	1.922	-	2.269	-	2.740	-	-	-
38.10	1.435	1.774	-	2.187	-	2.790	2.984	-	-	-	-
41.28	1.514	1.873	-	2.310	-	-	-	-	-	-	-
42.86	-	1.991	-	2.458	-	2.912	-	3.785	-	-	-
48.50	1.829	2.267	2.483	2.803	3.118	3.326	-	4.366	4.878	-	-
50.80	1.940	2.405	2.634	2.975	3.312	3.533	-	-	-	-	-
57.15	2.190	2.718	2.979	3.366	3.750	4.003	-	-	-	-	-
60.33	-	2.873	-	3.560	3.967	4.236	4.898	5.551	6.154	-	-
63.50	2.440	3.031	3.323	3.758	4.188	4.472	5.174	5.864	6.541	-	-
73.00	2.815	3.499	3.838	4.343	-	5.174	5.994	6.801	7.595	8.377	-
76.20	2.937	3.652	4.006	4.534	5.057	5.403	6.261	7.106	7.939	8.771	-
80.00	3.090	3.844	4.217	4.774	5.326	5.691	6.597	7.499	8.371	9.239	-
88.90	-	4.282	-	-	-	6.350	7.364	8.368	9.357	10.468	-
95.00	3.710	4.583	5.030	5.698	6.361	6.801	7.891	8.969	10.034	-	-
101.60	-	4.908	-	6.104	6.816	7.288	9.460	9.619	10.766	11.911	14.135
114.30	-	5.534	6.077	6.887	7.692	8.227	9.555	10.871	12.174	13.465	16.011
127.00	-	6.160	6.765	7.669	8.569	9.166	10.650	12.123	13.583	15.030	17.888
139.70	-	-	-	-	9.444	10.105	11.746	13.374	15.003	16.609	19.766
152.40	-	-	-	-	10.351	11.043	12.840	14.626	16.400	18.159	21.643
165.10	-	-	-	-	11.196	11.982	-	15.878	17.807	19.740	23.521

SQUARES

Outside diameter mm	Wall Thickness mm										
	1.6	2.0	2.2	2.5	2.8	3.0	3.5	4.0	4.5	5.0	6.0
12.70	0.564	-	-	-	-	-	-	-	-	-	-
15.90	0.726	-	-	-	-	-	-	-	-	-	-
19.10	0.938	1.153	-	-	-	-	-	-	-	-	-
25.40	1.198	1.478	-	1.817	-	2.129	-	-	-	-	-
31.80	1.514	1.873	-	2.310	-	-	-	-	-	-	-
38.10	1.829	2.267	2.483	2.803	3.118	3.326	-	-	-	-	-
44.50	2.190	2.718	2.979	3.366	3.750	4.003	-	-	-	-	-
50.80	2.440	3.031	3.323	3.758	4.188	4.472	5.174	5.864	6.541	-	-
63.50	3.090	3.844	4.217	4.774	5.326	5.691	6.597	7.499	8.371	-	-
76.20	3.710	4.583	5.030	5.698	6.361	6.801	7.891	8.969	10.034	-	-
80.00	-	4.908	-	6.104	6.816	7.288	9.460	9.619	-	-	-
90.00	-	5.534	6.077	6.887	7.692	8.227	9.555	10.871	12.174	13.465	16.011
101.60	-	6.160	6.765	7.669	8.569	9.166	10.650	12.231	13.583	15.030	17.888
120.00	-	-	-	-	10.351	11.043	12.840	14.626	16.400	18.159	-

STRUCTURAL TUBE

MILD STEEL DIMENTIONS AND MASS

RECTANGLES

Outside diameter mm	Wall Thickness mm										
	1.6	2.0	2.2	2.5	2.8	3.0	3.5	4.0	4.5	5.0	6.0
25.40	12.70	0.938	1.153	-	-	-	-	-	-	-	-
31.80	19.10	1.198	1.478	-	1.817	-	-	-	-	-	-
38.10	19.10	1.435	1.774	-	-	-	-	-	-	-	-
38.10	25.40	1.514	1.873	-	2.310	-	-	-	-	-	-
50.80	25.40	1.829	2.267	2.483	2.803	3.118	3.326	-	-	-	-
50.80	38.10	2.190	2.718	2.979	3.366	3.750	4.003	-	-	-	-
63.50	38.10	-	3.031	-	-	-	4.472	-	-	-	-
76.20	25.40	2.440	3.031	3.323	3.758	4.188	4.472	5.174	5.864	-	-
76.20	38.10	2.815	3.499	3.838	4.343	-	5.174	5.994	6.801	7.595	8.377
76.20	50.80	3.090	3.844	4.217	4.774	5.326	5.691	6.597	7.499	-	-
80.00	40.00	2.937	3.652	4.006	4.534	5.057	5.403	6.261	7.106	7.939	8.771
90.00	50.00	-	4.282	-	-	-	6.350	-	-	-	-
101.60	50.80	3.710	4.583	5.030	5.698	6.361	6.801	7.891	8.969	10.034	-
101.60	76.20	-	5.534	6.077	6.887	7.692	8.227	9.555	10.871	12.174	13.465
127.00	76.20	-	6.160	6.765	7.669	8.569	9.166	10.650	12.123	13.583	15.030
160.00	80.00	-	-	-	-	10.351	11.043	12.840	14.626	16.400	18.159

STRUCTURAL TUBE

SECTIONAL PROPERTIES

The sectional properties in the tables are based on the specified outside dimensions and wall thickness.
For square and rectangular sections the corner radii used in the calculations are:
R Internal = 1.5t and R External = 2.5t where t is the wall thickness.

The masses per unit length are calculated on the basis that steel has a density of 7.85 x 103 kg/mm3.

Torsional constants for most of the sections are available on request.

STEEL QUALITIES

SANS 657 Tubes shall be of steel chemical composition that complies with the following:

- Carbon content at 0.25% maximum
- Sulphur and Phosphorus at 0.05% each, maximum

DIMENSIONS

As per the sizes listed in the properties tables. These are largely in accordance with ISO/DIS4019 and other sizes can be supplied by agreement. The sizes listed are metric and we will be adopting these sizes as standard. All tubing is normally supplied in 6m lengths, but other lengths are available on application. Round tubing from 60mm and bigger; Square tubing from 50mm and bigger; Rectangular tubing 600 x 40mm and bigger. 3mm and thicker are available complying with S355MPa yield strength material.

TOLERANCES

Outside dimensions	± 1% with a minimum of ± 0.5mm
Wall Thickness	± 10% with a minimum of ± 0.2mm otside the weld area
Staightness	0.2% of total lenth

Other and closer tolerances can be supplied on request or on agreement.

WALL THICKNESS

The standard wall thicknesses available in the Republic of South Africa are:
- 1.6 - 2.0 - 2.2 - 2.5 - 2.8 - 3.0 - 3.5 - 4.0 - 4.5 - 5.0 - 6.0